

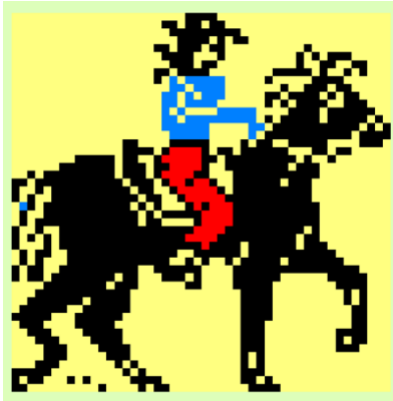


Cowboy Carl Game Design Document

By Alan Ramirez

Note : All sprites / designs are not final and subject to change.

Features



Player: Cowboy Carl

Sounds : Take Damage

Sprites : Running, Jumping, Shielding, Take Damage, Death



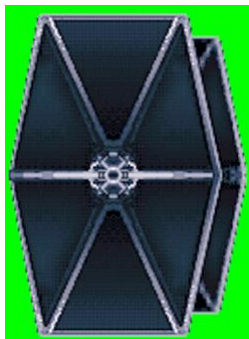
Boss: UFO

Sounds: Take Damage

Sprites: Flying,



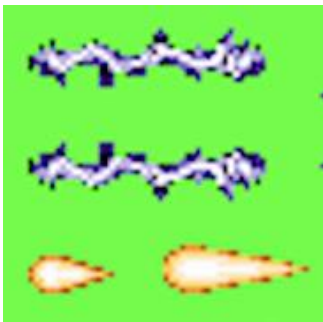
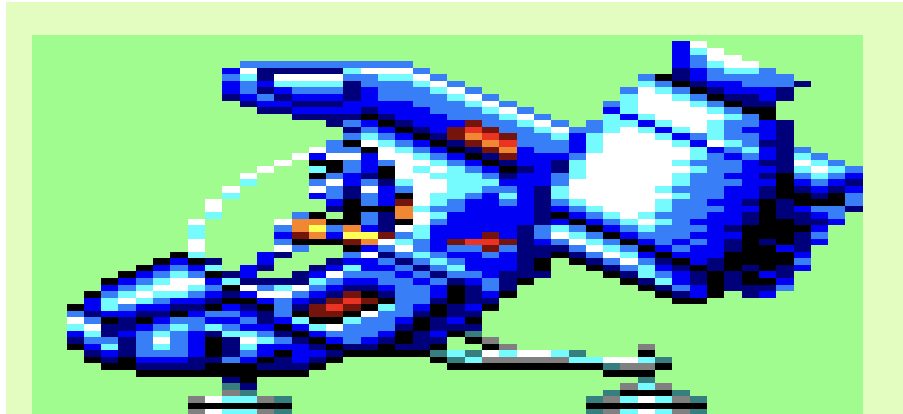
Enemy: Baller



Enemy: Hazard

Sprites: Level 1, level 2, level 3.

Enemy: Chopper

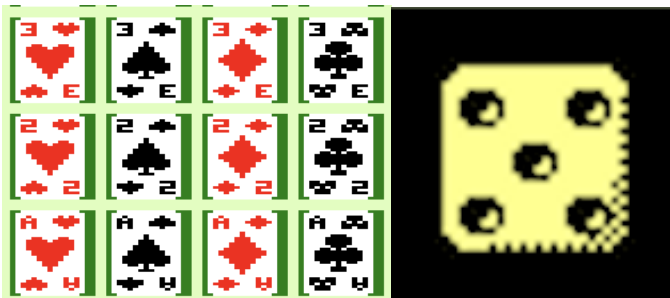


Enemy: Arrow .

Sprites: Level 1, level 2, level 3.

Objects

Card Bullet, Health Powerup, Ammo Powerup, Lasso, Boss Shield.

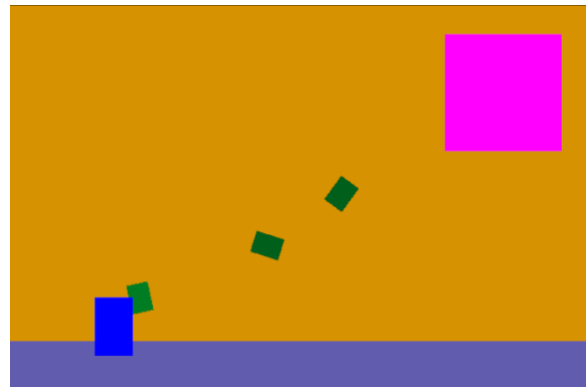
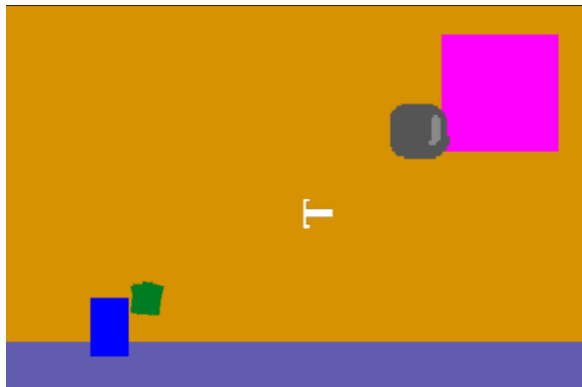


Gameplay



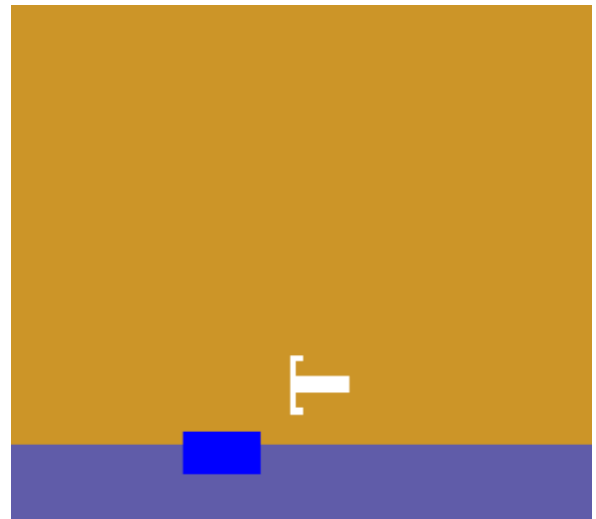
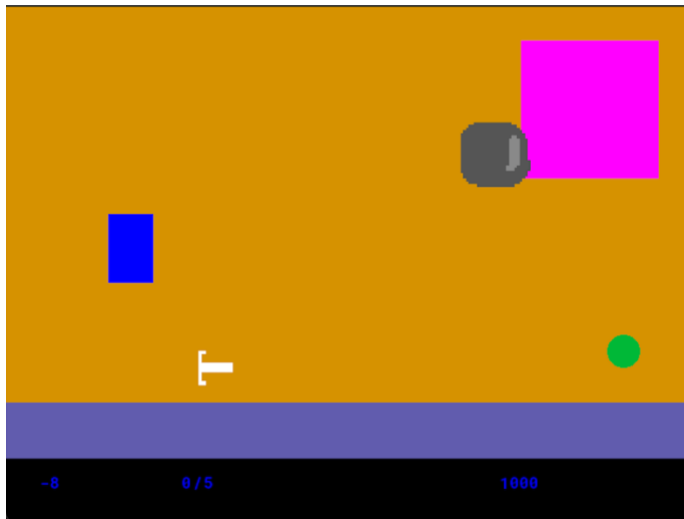
Auto-Run side-scroller. Shoot the Boss and other enemies while avoiding enemy attacks and hazards. Collect power ups and cosmetic items by jumping at the right time. The boss uses many tactics to avoid your attacks like using a shield. When boss health is zero the level will end.

Card Bullet attacks 5 at a time max. Displaying for 1 second and then fly towards target direction.



(IN-DEVELOPMENT SCREENSHOT) (Player is shown attacking target UFO.)

Strategic Decision-Making



(IN-DEVELOPMENT SCREENSHOT) (Player is shown jumping and shielding from enemy.)

At any time, you must choose whether to attack, jump, or shield. You cannot attack while jumping or shielding, and you can't shield while you are jumping. The enemies and objects are designed to test you on these actions. For example, the chopper drops an arrow on top of you and the only way to avoid it is to shield. If you jump while the chopper is above you, you will take damage.

If a collectable object spawns you must choose if you want to jump to grab it, because you might take damage if you try.

The level will only be complete if you attack the boss enough times. However, if all you do is attack the boss you will die because the enemies will be attacking you. So you have to choose when to choose a target, choose when to attack and when to defend.

When shooting, you can choose the diagonal or right direction only. Bullets take a second to fire, so you have to think ahead before you choose a direction to shoot.

Resource Management

At the start of the level, you can only shoot 2 bullet cards at a time, after shooting five, you have to wait for them to reload. Later on you can upgrade how many bullets you can shoot. This is to prevent the player from shooting in every direction all the time, that would make the game too easy.

Balance Between Features

It can be very difficult to encounter two or three types at once, so spawn times are adjusted to have good pacing. I call them enemy spawn clusters, a collection of spawn commands that might be fun for the player. For example I could spawn only 3 jump hazards in a row, so the player must press the spacebar 3 times with good rhythm. But for this to work, a chopper enemy must NOT exist, because it would cause the player to take damage while avoiding the hazard.

Who is in Control?

The player has control over the pace of the level because they have to shoot the boss to progress. Action systems like jumping and shielding give the player freedom of movement. Shielding is enabled instantly when player holds the button, and player can jump higher the longer they hold the jump button.

Variation

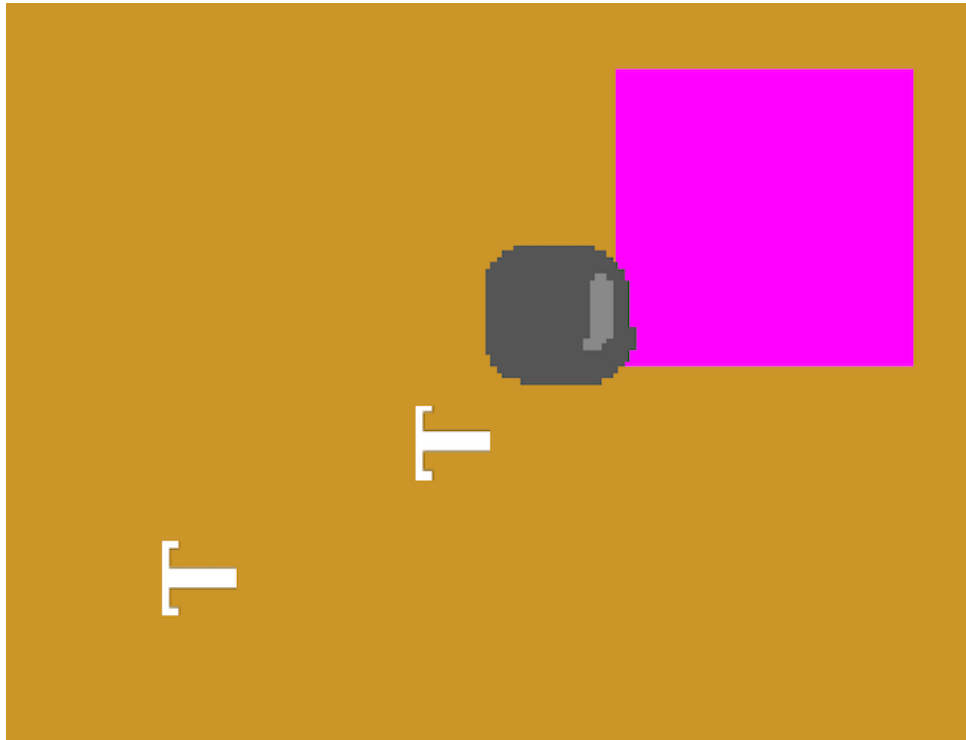


- The enemy spawn patterns give the gameplay variety.
- A power up like extra ammo will stack over the course of the level, so you may have significantly more bullets by the end.
- Lasso powerup will give the player more options for how to attack enemies.
- Collectables like hats will make the players more motivated to keep playing.
- Level design will change every level, enemy sprites and background sprites will change as well.
- The background will change as each level progresses.

AI

-The enemy spawn object will choose which enemies should appear to make the gameplay most fun, it will choose randomly from pre-set patterns. The number of attacks and attack speed will be chosen by the spawn object.

Enemy Behavior



(IN-DEVELOPMENT SCREENSHOT) (UFO shield active and Spawning Arrows)

UFO

The UFO boss enemy will always be at the top right of the screen. It will randomly spawn a shield to defend itself from your attacks. It will also randomly spawn arrows to attack you.

ARROW

An arrow is the most basic enemy type. It always moves toward the player, and it can be destroyed with a bullet card or avoided by jumping/shielding.

It can also be used like a bullet that each enemy can spawn. The enemy chooses a sprite/speed for its own arrow attack.

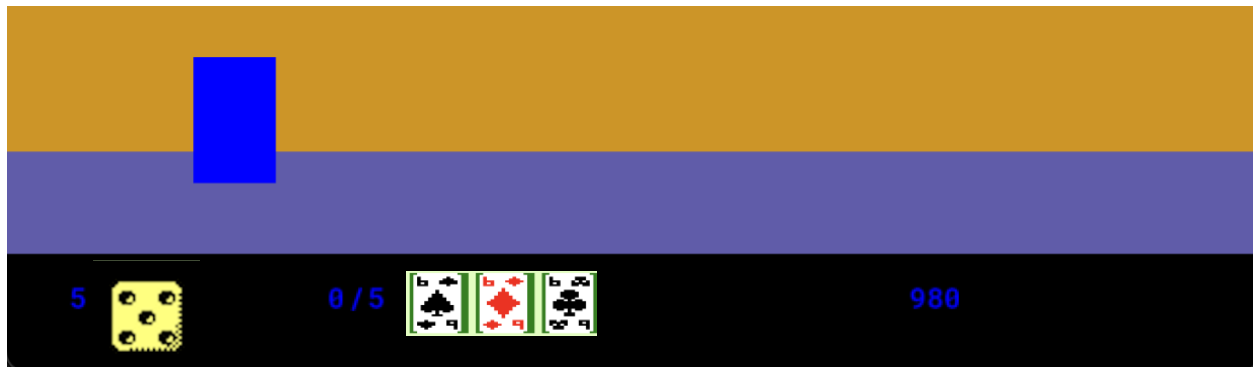
CHOPPER

The chopper cannot be attacked with a bullet card. It flies directly above the player drops a certain amount of arrows and then flies away. The player must avoid these attacks and wait for the chopper to leave. If the player has the lasso powerup, it can grab a chopper and launch it into the ufo or another enemy.

BALLER

The baller can be destroyed with bullet cards. It always spawns on the right side of the screen and shoots arrows infinitely at the player until the player destroys it. It can also be grabbed with the lasso powerup and launched into the ufo or other enemies.

UI



(IN-DEVELOPMENT SCREENSHOT) (Black Bar UI with Temp Number Values)

The UI will be displayed in a black rectangle at the bottom of the screen. All elements will be inside.

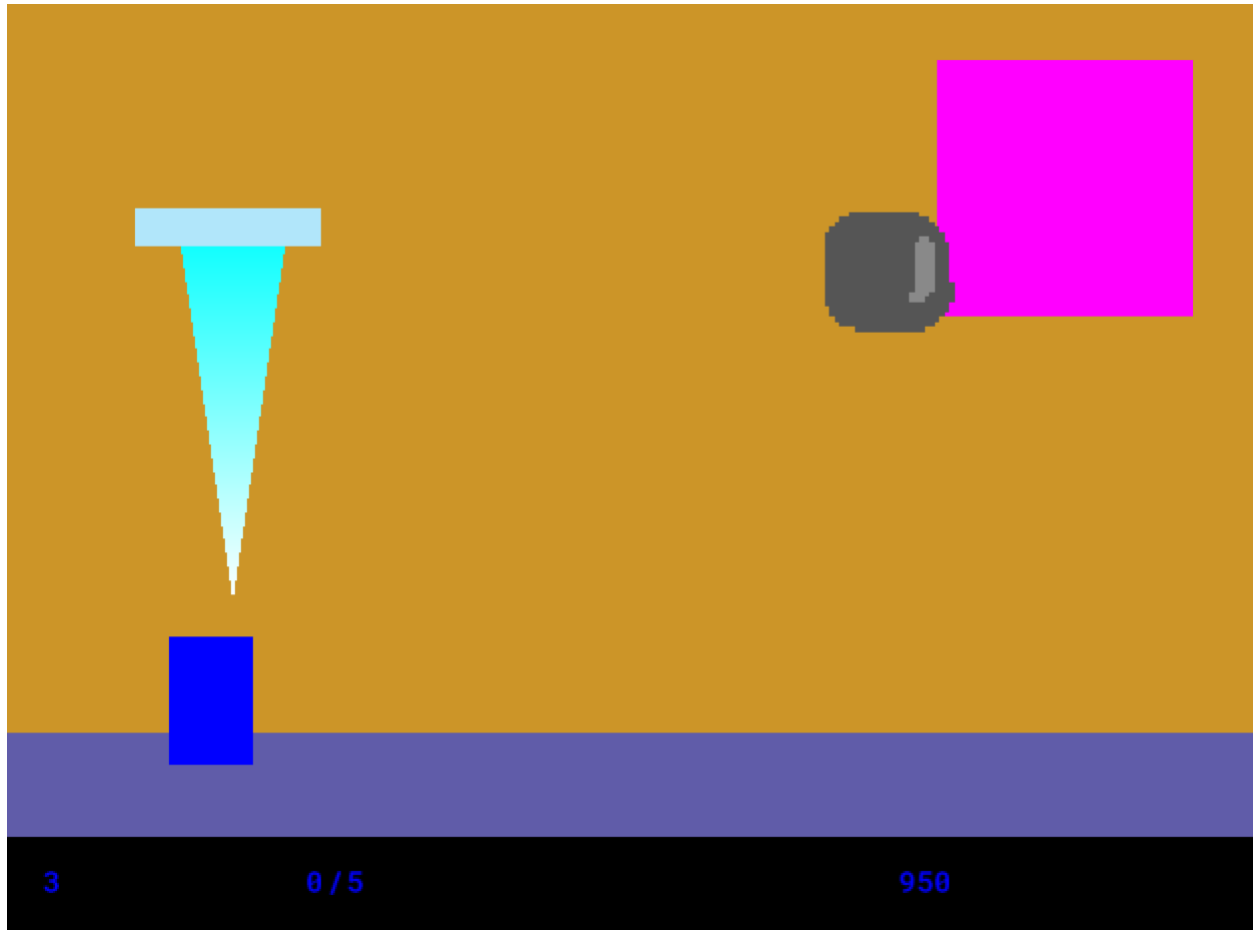
The Amount of lives will be displayed in numbers.

The ammo amount will be displayed as number of cards / total.

Lasso Cooldown timer will be displayed in the middle.

The boss health will be on the bottom right corner.

Goal



(IN-DEVELOPMENT SCREENSHOT) (Player is shown using Lasso Powerup.)

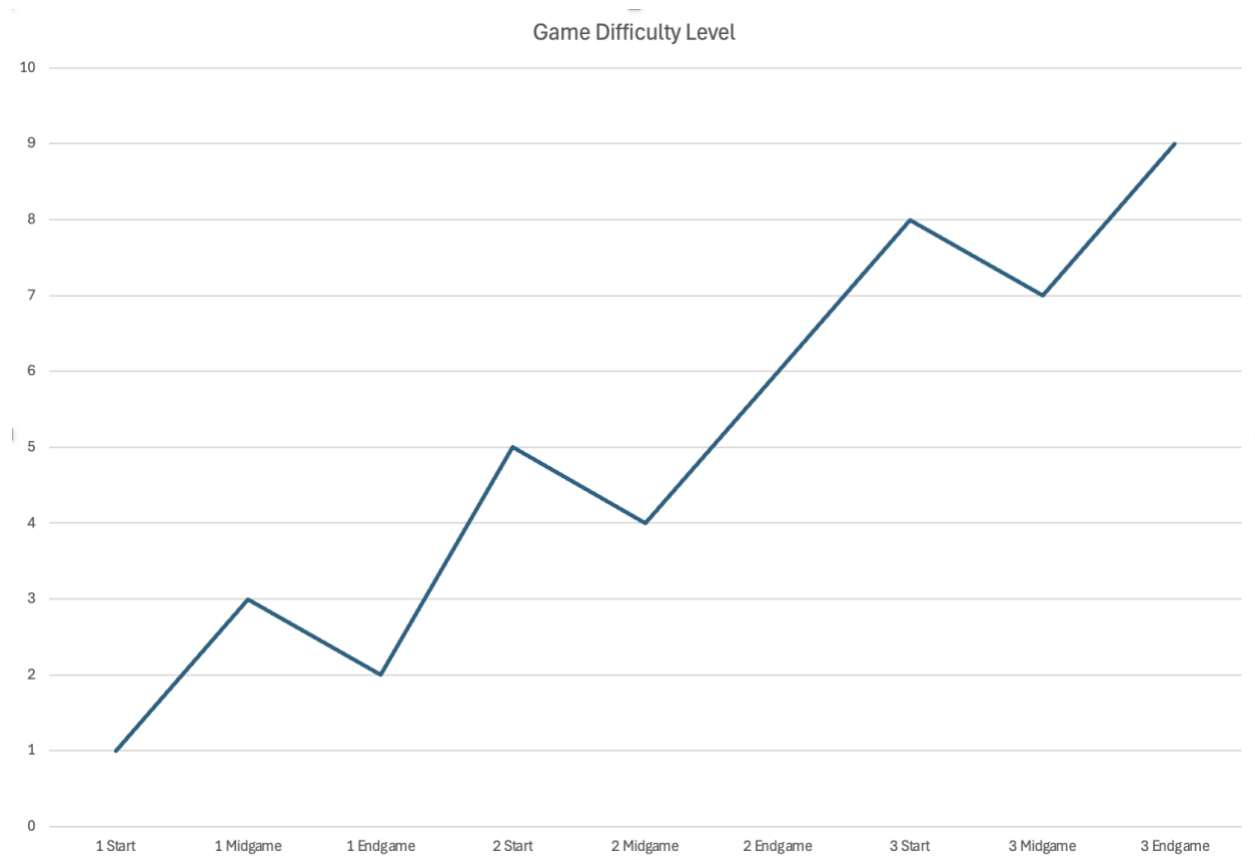
The goal of every level will be to defeat the Boss UFO. A side goal will be to collect powerups and do cool moves with the lasso.

- Difficulty graph from start to end of the game, how

each level ramps up the difficulty using the various gameplay elements

- Hierarchy charts that show the inheritance between various parent child objects
- Excel spread sheet that lists the timeline of completing the various tasks by last day of the class

Difficulty



The game will start very easy so players can learn the controls and enemy attacks.

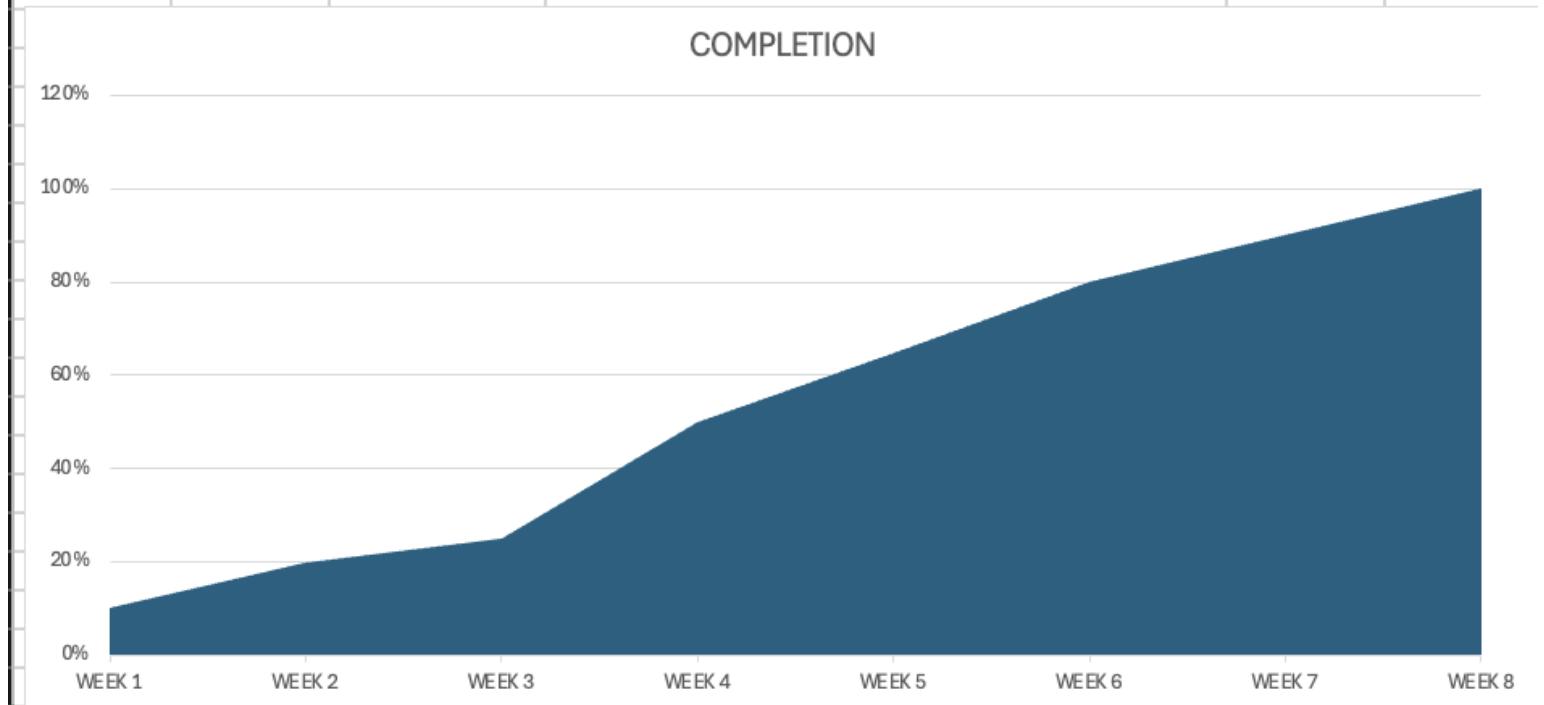
At the middle of the level it will get harder, but as the player collects powerups, things will get easier.

Each new level will be significantly more difficult due to enemy speed and spawn frequency.

In level 2 and 3 the middle will get easier because of power ups, but when they boss begins to reach lower health, spawn frequency and speed will make the level harder at the end.

Production Timeline

WEEK #	DATE	COMPLETION	DESCRIPTION	Column1	Column2
WEEK 1	23-Mar-26	10%	Initial Idea, Begin Prototype (player controls, scrolling background)		
WEEK 2	30-Mar-26	20%	Gameplay Prototype (Create Enemies)		
WEEK 3	3-Apr-26	25%	PlayTest Prototype / New Ideas (Create New Enemies, Power ups)		
WEEK 4	6-Apr-26	50%	Begin Actual Project with cleaner code (Create Spawns and Levels)		
WEEK 5	13-Apr-26	65%	Create GDD And playtest game. Adjust difficulty, Add UI		
WEEK 6	20-Apr-26	80%	Add sprites/animations/backgrounds, adjust mechanics. Playtest		
WEEK 7	27-Apr-26	90%	Add effects/ title screens and level screens, Polish		
WEEK 8	4-May-26	100%	Add Sound and music. Polish		



Inheritance Chart

